

HAT-P-37b

R-2570

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-37_210625 · created 2026-08-26T23:52:05+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459390.7036 +/- 0.0049 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.171 +/- 0.016
Transit depth (Rp/Rs)^2	2.91 +/- 0.56 [%]
Orbital Inclination (inc)	84.54 +/- 1.34
Airmass coefficient 1 (a1)	0.79 +/- 0.015
Airmass coefficient 2 (a2)	0.157 +/- 0.032
Scatter in the residuals of the lightcurve fit is	1.9 %
Best Comparison Star	#3 - [159, 479]
Optimal Aperture	2.59
Optimal Annulus	7.03
Transit Duration (day)	0.068 +/- 0.019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.171 $\pm$ 0.016 vs 0.1394 (deviation 1.75 σ)
Duration fit vs published	0.068 d vs 0.0983 d (deviation 1.41 σ)
Mid-time O-C	-17.9 min
Depth SNR	9.4
Residual scatter	1.9 %

Light curve

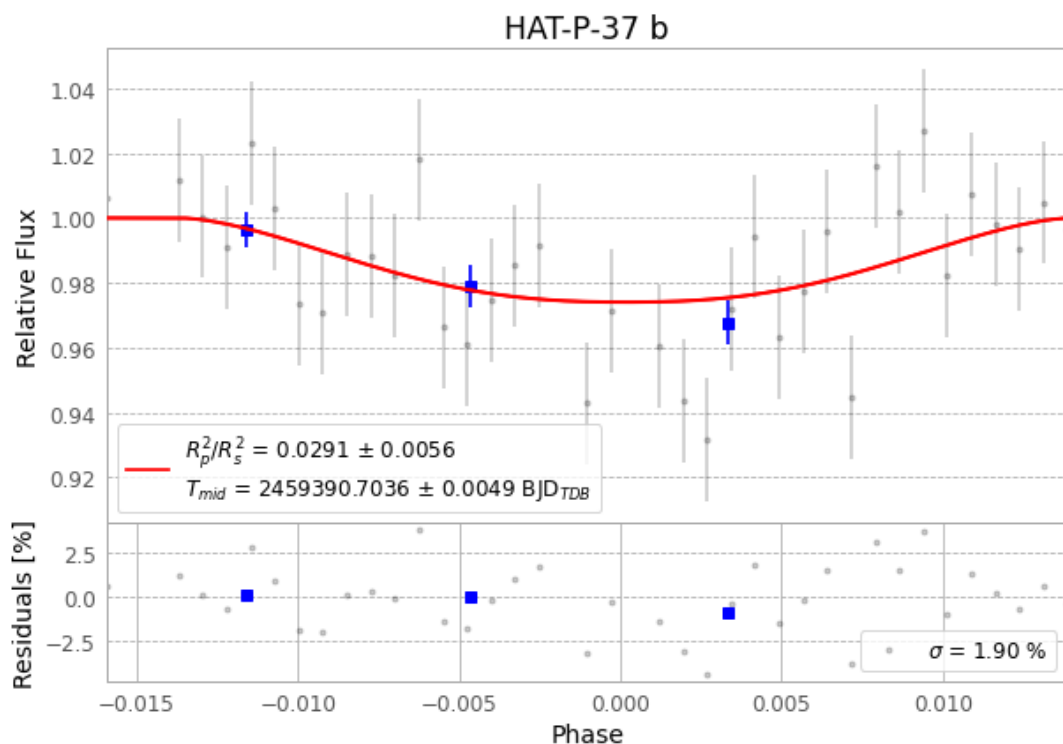


Figure 1 — FinalLightCurve_HAT-P-37 b_2021-06-24.png (SHA-256 f925e1d02fd57000...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [284, 264], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 28c530e694572656...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

39 FITS frames (26 MB), HATP-37210625034514.FITS → HATP-37210625054512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-37-210625.log (8b4cf27859f2...), FinalLightCurve_HAT-P-37 b_2021-06-24.png (f925e1d02fd5...), FinalLightCurve_HAT-P-37 b_2021-06-24.csv (8d8404b69535...)

Compute resources

Wall time 118.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-26 23:52Z.